

AgileTech Studio

Engineering Reference Document | Owner: Zainab Qureshi, Head of Engineering | Updated July 2026

System Architecture — NovaCart & MedLink

1. NovaCart Architecture

NovaCart is a multi-tenant e-commerce platform built for RetailPro Inc. It follows a modular monolith pattern on the backend, deployed as containerized services on AWS.

Layer	Technology	Notes
API Layer	Python / FastAPI	REST API, OpenAPI schema auto-generated
Frontend	React + TypeScript	SPA, server-side rendering disabled for this phase
Database	PostgreSQL (AWS RDS)	Primary transactional store; read replicas planned for Q4 2026
Caching	Redis	Session cache and checkout cart state
Hosting	AWS ECS (Fargate)	Auto-scaling based on request load
CI/CD	GitHub Actions	Deploys to staging on merge to main, production on tagged release

Ownership: Ayesha Khan (Team Lead) owns the API and frontend architecture; Hamza Sheikh owns backend services; Mahnoor Butt owns frontend component architecture.

2. MedLink Architecture

MedLink is a cross-platform mobile health app for HealthBridge Ltd., built with a document-oriented data model to accommodate flexible, evolving patient record schemas.

Layer	Technology	Notes
API Layer	Node.js (Express)	REST + WebSocket for real-time notifications
Mobile Client	React Native	Single codebase for iOS and Android
Database	MongoDB Atlas	Patient records, appointment data
Hosting	AWS (EC2 + S3)	S3 for encrypted document/image storage
Security	Field-level encryption	Required for patient-identifiable data per HealthBridge contract

Ownership: Danish Iqbal (Team Lead) owns mobile and API architecture; Usman Tariq supports mobile feature development; Noor Fatima owns QA and test architecture, including end-to-end test coverage for patient-data workflows.

3. Cross-Project Standards

Both systems follow the API design conventions in API_Standards.docx and the deployment approval process in Deployment_Processes.pdf, which requires DevOps sign-off from Ali Raza or Waleed Anjum before any production release.